Estuti Shukla

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EXPERIENCE

Graduate Student Researcher Aug. 2022 – Present
The Pennsylvania State University — Advisor: Prof. David Radice *University Park, PA*

- Developing scientific machine learning methods to solve fundamental problems in general relativity
- Performing high-resolution simulations of spinning compact binary coalescences using `GR-Athena++` and GPU-capable `AthenaK`, scalable numerical relativity codes for high-performance computing

Visiting Student Researcher Sept. 2021 – Aug. 2022
ICTS–TIFR Bangalore — Advisor: Prof. Prayush Kumar *Bangalore, India*

- Studied properties of dynamical horizons using numerical relativity simulations with the `SpEC`, a pseudo-spectral high-performance computing code for numerical relativity

Research Intern Oct. 2020 – Aug. 2022
Johns Hopkins University — Advisor: Dr. Vishal Baibhav & Prof. Emanuele Berti *Remote*

- Investigated nonlinearities of quasi-normal modes associated with ringdown of binary black hole mergers

Summer Research Intern Jun. – Sept. 2020
University of Massachusetts Dartmouth — Advisor: Prof. Gaurav Khanna & Prof. Scott Field *Remote*

- Developed calculations of Bondi mass and recoil kick velocities from black-hole perturbation theory waveforms
- Contributed to early calculations underlying the open-source `gw_remnant` package; acknowledged in Phys. Rev. D 108, 064048 (2023)

EDUCATION

The Pennsylvania State University State College, USA
Ph.D. in Physics *Aug. 2022 – Present*

- Advisor: Prof. David Radice
- Research: AI4Science, numerical relativity, black hole perturbation theory

Indian Institute of Science Education and Research Kolkata Kolkata, India
B.S.–M.S. in Physics — GPA (major): 9.03/10.00 *2016 – 2021*

- Master's Thesis: *Static Stars in Cosmological Backgrounds*, advisor Prof. Rajesh Nayak
- DST Inspire Fellowship (Govt. of India)

RESEARCH INTERESTS

Scientific machine learning, numerical relativity, high performance computing, and gravitational wave physics.

PUBLICATIONS

[1] **E. Shukla**, D. Radice, R. Gamba, and S. Bernuzzi, "Gauge Fixing with Neural Networks," arXiv (2026).

[2] **E. Shukla**, A. Suriyanarayanan, D. Radice, R. Maulik, and S. Bernuzzi, "Puncture Evolution with Physics-Informed Neural Networks,", submitted to PRX Intelligence, arXiv:2610.06776 (2026).

[3] R. Gamba, D. Chiaramello, **E. Shukla**, and S. Albanesi, "Spin the black circle II: tidal heating and torquing of a rotating black hole by a test mass on generic orbits," accepted in *Phys. Rev. D*, arXiv:2603.28982 (2026).

[4] **E. Shukla**, A. Rashti, R. Gamba, D. Radice, and K. Chandra, "`GR-Athena++` simulations of spinning binary black hole mergers," *Class. Quant. Grav.* **43**, 065019 (2026).

[5] R. Perna, O. Gottlieb, **E. Shukla**, and D. Radice, "Connecting GRBs from binary neutron star mergers to nuclear properties of neutron stars," *Phys. Rev. D* **111**, 063015 (2025).

[6] M. H.-Y. Cheung, V. Baibhav, E. Berti, V. Cardoso, G. Carullo, R. Cotesta, W. Del Pozzo, F. Duque, T. Helfer, **E. Shukla**, and K. W.-K. Wong, "Nonlinear effects in black hole ringdown," *Phys. Rev. Lett.* **130**, 081401 (2023).

TALKS & CONFERENCES

Invited Talks

Strong Gravity Seminar, Penn State, PA	Sept. 2026
AstroChat Talk, UC Berkeley, CA	July 2026
Physics Colloquium, UMass Dartmouth, MA *(virtual)*	Sept. 2020

Contributed Talks

ICDS Symposium, Penn State, PA *(Poster)*	Oct. 2026
AstroAI Workshop 2026, Harvard University, MA	Jun. 2026
26th Eastern Gravity Meeting, Cornell University, Ithaca, NY	May 2026
AI for Gravitational Waves Workshop, CERN, Geneva *(Poster)*	May 2026
Neighbourhood Workshop, Penn State, PA	Apr. 2026
35th Midwest Relativity Meeting, U. Guelph, Canada	Nov. 2025
ICDS Symposium, Penn State, PA *(Poster)*	Sept. 2025
Scientific Machine Learning for GW Astronomy, ICERM, RI *(Poster)*	Jun. 2025
IGC Alumni Event, Penn State, PA *(Poster)*	May 2025
APS Global Physics Summit 2025, Denver, CO	Mar. 2025
Primordial Universe and Gravity (PUG) Seminar, Penn State, PA	Apr. 2024

Schools & Workshops

IAIFI Summer School, UMass Boston, MA *(virtual)*	Aug. 2026
Deep Learning for Science School, Berkeley, CA	Jul. 2026
AthenaK Summer School, Penn State, PA	Jul. 2026
Neighbourhood Workshop, Penn State, PA	2023 – 2025
Simulating Extreme Spacetimes with SpEC and SpECTRE, ICERM, RI	Aug. 2024
DTP TALENT Summer School on Nuclear Astrophysics, Trento, Italy	Jul.-Aug. 2024
N3AS Summer School in Multi-messenger Astrophysics, UC Santa Cruz, CA	Jul. 2023
ICTS School on Physics of Early Universe, ICTS-TIFR, India	Jan. 2022
Finesse Interferometer Modelling Hackathon, IUCAA, India *(winner)*	Mar. 2020
National Science (Vijyoshi) Camp, DST, Gov. of India, IISER Kolkata	Dec. 2016

AWARDS & HONORS

APS DGRAV Student Travel Award, APS Global Summit	2025
EUSTIPEN Travel Award, Trento, Italy	2024
Homer F. Braddock Scholarship, Penn State University	2023
DST Inspire Scholarship, Government of India	2016-2021

TECHNICAL SKILLS

Programming (Advanced): Python, PyTorch, LaTeX, C/C++, Mathematica
Programming (Intermediate): R, MATLAB
Numerical Relativity Codes: AthenaK, GR-Athena++, SpEC
Tools: Git, GNUplot, Origin, Linux, Claude Code

TEACHING AND MENTORSHIP

Teaching Assistant, Penn State University 2022-2024

- PHYS 212 General Physics: Electricity and Magnetism
- PHYS 250 Introductory Physics 1

Teaching Assistant, IISER Kolkata 2020-2021

- General Theory of Relativity & Cosmology (w/ Prof. Rajesh Nayak)
- Intermediate Quantum Mechanics (w/ Prof. Soumitro Banerjee)

Undergraduate Student Mentor

- Ben Jhonson, Penn State University
- Patrick A Bush, San Diego State University (graduate student at OSU)

PROFESSIONAL ACTIVITIES

Journal Referee

- Classical and Quantum Gravity

Member

- LIGO-Virgo-KAGRA Consortium (2022-)
- APS DGRAV, FGSA, FIP (2023-)
- Simulating eXtreme Spacetimes Collaboration (2021-2022)

OUTREACH AND SERVICE

Lecture on Gravitational Waves, State College Area High School	Apr 2026
Professional Development Chair, Physics and Astronomy for Women+ (PAW+), Penn State	2024-2027
Physics Graduate Student Representative, Penn State	2025-2026
Board Member, Physics Graduate Student Association (PGSA), Penn State	2025-2026
Delegate for ECoS, Graduate & Professional Student Association (GPSA), Penn State	2025-2026
Chair of Academic Affairs Committee, GPSA, Penn State	2025-2026
Founder of IKQRAAR, official LGBTQ+ support platform, IISER Kolkata	2021-2022
Founder of Trekking & Adventure Club, IISER Kolkata	2020-2021
Founding Member of Student Alumni Cell, IISER Kolkata	2019-2020
Sponsorship Head of Inquivesta, college science festival, IISER Kolkata	2019-2020

REFERENCES

Prof. David Radice
Associate Professor at the Departments of Physics and Astronomy & Astrophysics
Pennsylvania State University
email: dur566@psu.edu
Prof. Romit Maulik
Assistant Professor of Mechanical Engineering
Purdue University
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Prof. B.S. Sathyaprakash
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